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AES standard for digital audio - Audio-embedded metadata - Part 5: EBU loudness, true-peak, and downmix

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AES standard for digital audio - Audio-embedded metadata - Part 5: EBU loudness, true-peak, and downmix

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Abstract

AES41 provides for the carriage of audio metadata by embedding it in the audio samples themselves. This tightly associates the meta-data with the audio, yet makes it fragile so that changes to the audio will invalidate the meta-data. Several metadata sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio to signal loudness and true peak meta-data as used in EBU Technical Recommendation R128. A method of carrying this data is described in Part 1 of this Standard.

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Foreword

This foreword is not part of AES41-5-2012 *AES standard for digital audio - Audio-embedded metadata - Part 5: EBU loudness, true-peak, and downmix*

This document describes a set of data that may be conveyed according to the method described in Part 1 of this Standard.

As predicted in the foreword to AES41-2000, digital compression techniques now dominate the broadcast television environment. In addition to the problems foreseen relating to cascaded compression, new problems have arisen because of the use of loudness control and surround sound with those digital compression techniques.

Metadata within the compressed audio bit-stream is used to control loudness and the mixing down of multi-channel surround sound to two-channel stereo. These metadata are usually known by terms such as "dialnorm", "prog_ref_level", and "down-mix coefficients".

Whilst this might seem unrelated to the original scope of AES41, dealing with bit allocations and scale factors, it is simply another form of data that can affect a later encoding of the audio: this time it is more macroscopic than microscopic.

The metadata is lost when the bit-stream is uncompressed unless provision is made to transport it or store it somewhere. Existing methods rely on non-audio mechanisms to convey the metadata alongside the audio, for example a serial data link like RS-422 and serial digital video SMPTE 259M, or a "chunk" in an audio file (for metadata that does not change).

This part extends AES41 to include a data format for carrying loudness metadata as defined by the EBU following the work of the "PLOUD" group, as well as downmix metadata, with the uncompressed PCM using the same transport mechanism as before. The metadata can therefore be carried in the audio to which it relates.

The draft of this document was developed by a writing group whose primary author was Andrew Mason.

John Grant
Chair, working group SC-02-02
2012-03

Note on normative language

In AES standards documents, sentences containing the word "shall" are requirements for compliance with the document. Sentences containing the verb "should" are strong suggestions (recommendations). Sentences giving permission use the verb "may". Sentences expressing a possibility use the verb "can".

AES standard for digital audio - Audio-embedded metadata - Part 5: EBU loudness, true-peak, and downmix

0 Introduction

AES41 provides for the carriage of audio meta-data by embedding it in the audio samples themselves. This tightly associates the meta-data with the audio, yet makes it fragile so that changes to the audio will invalidate the meta-data. Several meta-data sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio to signal loudness and true peak meta-data as used in EBU Technical Recommendation R128. This part should be read in conjunction with part 1.

0.1 Rationale for part 5 of this standard

Unwanted loudness variations in broadcast audio have historically been the source of many audience complaints. Broadcasters have adopted numerous techniques to address the problem - one of which involves indicating the long-term average loudness of the audio.

The EBU studied the needs of audio signal levels in production, distribution and transmission of broadcast programmes. It formed the opinion that an audio-leveling paradigm was needed based on loudness measurement according to ITU-R BS.1770. This is described in EBU Technical Recommendation R 128, and supporting Technical Documents.

When the PCM signals form part of a 5-channel surround sound signal it may be necessary to include down-mix coefficients to allow later mixing of the center and surround channels into the front left and front right to form a two-channel stereo signal.

1 Scope

This document describes a format for the data to be transmitted to convey loudness and true peak metadata as defined by the EBU following the work of the "PLOUD" group. This Part assumes that the transmission mechanism according to Part 1 of this Standard is used.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

AES41-1-2012, *AES standard for digital audio - Audio-embedded metadata - Part 1: General*, Audio Engineering Society, New York, NY., US.

3 Definitions and abbreviations

3.1 Definitions

3.1.1

LUFS

units of subjective loudness - "loudness units relative to full scale". The EBU uses units of 'LUFS' for subjective loudness measurements made using the measurement method described in ITU-R BS.1770. 'LUFS' is equivalent to 'LKFS', as used within ITU-R BS.1770. 'LUFS' follows the naming convention of units in compliance with ISO 80000-8, whilst 'LKFS' does not.

3.2 Symbols

The mathematical operators used in this standard are similar to those used in the C programming language. The bitwise operators are defined assuming 2's-complement representation of integers. Numbering and counting loops generally begin from zero.

3.2.1 Arithmetic operators

3.2.1.1

+

addition

3.2.1.2

++

increment

3.2.1.3

--

decrement

3.2.1.4

*

multiplication

3.2.1.5

%

modulus operator, defined only for positive numbers

3.2.2 Relational operators

3.2.2.1

<

less than

3.2.2.2

==

equal to

3.2.2.3

!=

not equal to

3.2.3 Bitwise operators

&
AND

3.2.4 Assignment

=
assignment operator

3.3 Abbreviations

The following mnemonics are defined to describe the different data types used in the coded bit stream.

3.3.1

bslbf

bit string, left bit first, where "left" is the order in which bit strings are written in this standard.

NOTE Bit strings are written as a string of logic 1's and logic 0's within single quote marks, for example, '1000 0001'. Blanks within a bit string are for ease of reading and have no significance.

3.3.2

simsbf

signed integer, most significant bit first

3.3.3

uimsbf

unsigned integer, most significant bit first

4 EBU loudness, true-peak, and downmix metadata

4.1 AES41 metadata type

The presence of this data in the stream shall be indicated by **type**=6₁₀, as described in Part 1 of this standard.

4.2 Syntax

The syntax of **data()** is specified in the box below in accordance with Part 1.

Syntax	No. of bits	Mnemonic
data()		
{		
if(type==EBU loudness metadata){		
LoudnessValue	16	simsbf
LoudnessRange	16	simsbf
MaxTruePeakLevel	16	simsbf
MaxMomentaryLoudness	16	simsbf
MaxShortTermLoudness	16	simsbf
MPEG2_mix_level_on	1	bslbf
Center_mix_level_value	3	bslbf
Surround_mix_level_value	3	bslbf
}		
for(p=data_bits%8; p != 0; p--)		
padding_bit	1	
}		

4.3 Semantics

The loudness parameters are represented by integer values, but preserve a precision of 2 decimal places by being multiplied by 100 before being rounded. The rounding function that shall be used is defined as follows:

integer representation, $\text{round}(x) = \text{integer part of } (x + \text{sgn}(x) \cdot 0.5)$

where x is the loudness parameter to be represented, multiplied by 100, and $\text{sgn}(x)$ is the signum function (-1 for $x < 0$; 0 for $x = 0$; +1 for $x > 0$). This rounding method is commonly referred to as, "round to nearest, ties away from zero". Examples are shown in table 1.

Table 1 examples of rounding and scaling method

Parameter value	Integer representation
12,766	1277
12,765	1277
12,764	1276
-22,644	-2264
-22,645	-2265
-22,646	-2265

LoudnessValue	A 16-bit signed integer, equal to $\text{round}(100 \times \text{the Integrated Loudness Value of the signal in LUFS})$	
LoudnessRange	16-bit signed integer, equal to $\text{round}(100 \times \text{the Loudness Range of the signal in LU})$	
MaxTruePeakLevel	16-bit signed integer, equal to $\text{round}(100 \times \text{the Maximum True Peak Value of the signal in dBFS})$	
MaxMomentaryLoudness	16-bit signed integer, equal to $\text{round}(100 \times \text{the highest value of the Momentary Loudness Level of the signal in LUFS})$	
MaxShortTermLoudness	16-bit signed integer, equal to $\text{round}(100 \times \text{the highest value of the Short-term Loudness Level of the signal in LUFS})$	
MPEG2_mix_level_on	Indicates whether the following mix_level_value parameters are valid	
	0	not valid, value should be ignored
	1	valid
Center_mix_level_value	Indicates the level at which the center channel shall be mixed into front left and front right. Corresponding multiplication factors are shown in table 2	
Surround_mix_level_value	Indicates the level at which the left surround and right surround channels shall be mixed into the front left and front right channels respectively. Corresponding multiplication factors are shown in table 2	
padding_bit	indicates bits that are added to bring the total number of bits in the data to a multiple of 8 to match the length field.	

Note 1: The loudness terms are explained more fully in EBU Recommendation R 128, EBU Tech Doc 3341 and EBU Tech Doc 3342, supported by EBU Tech Doc 3343 and EBU Tech Doc 3344.

Note 2: Not all loudness metadata fields will be measured or required in all circumstances. The value -1 ('0111 1111 1111 1111' as a bit string) shall be used to signal "field not indicated". Parameters signalled in this way shall be ignored.

Values of loudness level or true peak that are signalled outside the range -99.9 to +99.9 are invalid and shall be ignored. Values of loudness range that are signalled outside the range 0.00 to +99.9 are invalid and shall be ignored.

Table 2 - `mix_level_value` multiplication factors

Bit string	Multiplication factor
000	1,000
001	0,841
010	0,707
011	0,596
100	0,500
101	0,422
110	0.355
111	0,000

5 Data alignment

When the data set described in this Part is embedded in an audio signal, the first bit of the `data_frame` shall be embedded in the first audio sample to which the data applies. The repetition rate of an unchanging data set may be chosen to suit the requirements for "break-in" at arbitrary time to the audio signal of the application. At one extreme, this would be signaling only once, at the beginning of the stream. However, it is recommended that the repetition rate shall be the same as the frame rate of any associated video signal; if there is no associated video, the repetition rate shall be that of the predominant video frame rate in the environment.

EXAMPLE: in many territories, including Europe, the video frame rate is 25 Hz. In other countries, including the US and Japan, the video frame rate is 30/1,001 Hz (nominally 29,97Hz).

Annex A Bibliography

ITU-R Recommendation BS.1770-1, Algorithms to measure audio programme loudness and true-peak audio level, International Telecommunications Union, Geneva, Switzerland.

EBU Recommendation R 128, *Loudness normalisation and permitted maximum level of audio signals*, European Broadcasting Union, Geneva, Switzerland.

EBU Technical Document 3341, *Loudness Metering: 'EBU Mode' metering to supplement loudness normalisation in accordance with EBU R 128*, European Broadcasting Union, Geneva, Switzerland.

EBU Technical Document 3342, *Loudness Range: A descriptor to supplement loudness normalisation in accordance with EBU R 128*, European Broadcasting Union, Geneva, Switzerland.

EBU Technical Document 3343, *Practical guidelines for Production and Implementation in accordance with EBU R 128*, European Broadcasting Union, Geneva, Switzerland.

EBU Technical Document 3344, *Practical guidelines for distribution systems in accordance with EBU R 128*, European Broadcasting Union, Geneva, Switzerland.